GEN_DISTANCE

Overview

gen_distance allows users to generate single-bit pulses with a constraint-random distance between them. Two parameters, “min_clocks” and “max_clocks”, constrain the random generation of a pulse-to-pulse timing interval.

Syntax

```
gen_distance #(min_clocks, max_clocks) pulse_gen (clk, rst_n,
data_out);
```

Parameters	
Min_clocks	Minimum limit of inter-pulse distance
Max_clocks	Maximum limit of inter-pulse distance
Interface	
clk	Sampling clock for data
rst_n	Active-low reset
data_out	Output 1-bit data signal represents a generated pulse

Verilog example

```
module test;

reg clk;
wire data_out;
integer i;

always #5 clk <= ~clk;

gen_distance #(12, 15) dist (clk, 1'b1, data_out);

initial begin
  $dumpvars;
  $monitor ("data_out = %b, time = %0t", data_out, $time);
  clk = 1;
  #1000 $finish;
end

endmodule
```